

Lesson 13: Tests About a Population Mean and Proportion

BSTA 551: Statistical Inference

Jessica Minnier

2026-02-09

Lesson 13: Tests About a Population Mean and Proportion

Review: Where We Left Off

Lesson 12: Introduction to Hypothesis Testing

- Null and alternative hypotheses (H_0 vs H_a)
- Test statistics and rejection regions
- Type I error (α) and Type II error (β)
- Power = $1 - \beta$
- Connection between CIs and hypothesis tests

Review: Where We Left Off

Lesson 12: Introduction to Hypothesis Testing

- Null and alternative hypotheses (H_0 vs H_a)
- Test statistics and rejection regions
- Type I error (α) and Type II error (β)
- Power = $1 - \beta$
- Connection between CIs and hypothesis tests

Today's Goals:

1. Apply the one-sample z test for μ (σ known)
2. Develop the one-sample t test for μ (σ unknown)
3. Understand when to use z vs t tests
4. Conduct tests about a population proportion p
5. Implement all tests in R with medical research examples

Roadmap for Today

Topic	Assumption	Test Statistic	Textbook
z test for μ	σ known	$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$	Devore 9.2
t test for μ	σ unknown	$T = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$	Devore 9.2
z test for p	Large n	$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	Devore 9.3

Roadmap for Today

Topic	Assumption	Test Statistic	Textbook
z test for μ	σ known	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	Devore 9.2
t test for μ	σ unknown	$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	Devore 9.2
z test for p	Large n	$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	Devore 9.3

Connection to CIs

Each test today has a corresponding confidence interval from Lessons 7-8. The CI contains all values of the parameter that would **not** be rejected by the corresponding test.

The One-Sample z Test (Devore 9.2)

Setup and Assumptions

Assumptions:

1. $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution
2. The population standard deviation σ is **known**

Setup and Assumptions

Assumptions:

1. $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution
2. The population standard deviation σ is **known**

Null hypothesis: $H_0 : \mu = \mu_0$

Test statistic:

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Under H_0 : $Z \sim N(0, 1)$

The One-Sample z Test: Rejection Regions

! One-Sample z Test (Devore 9.2)

Alternative	Rejection Region	Type
$H_a : \mu > \mu_0$	$z \geq z_\alpha$	Upper-tailed
$H_a : \mu < \mu_0$	$z \leq -z_\alpha$	Lower-tailed
$H_a : \mu \neq \mu_0$	$ z \geq z_{\alpha/2}$	Two-tailed

The One-Sample z Test: Rejection Regions

! One-Sample z Test (Devore 9.2)

Alternative	Rejection Region	Type
$H_a : \mu > \mu_0$	$z \geq z_\alpha$	Upper-tailed
$H_a : \mu < \mu_0$	$z \leq -z_\alpha$	Lower-tailed
$H_a : \mu \neq \mu_0$	$ z \geq z_{\alpha/2}$	Two-tailed

Key idea: The standardized test statistic Z measures how many standard errors $\bar{x}$ is from μ_0 . Rejection occurs when this distance is “too large” in the direction specified by H_a .

Seven-Step Procedure (Devore 9.2)

Recommended Steps for Any Hypothesis Test

1. **Identify** the parameter of interest and describe it in context
2. **State** the null and alternative hypotheses
3. **Check** assumptions for the test procedure
4. **Write** the formula for the test statistic
5. **State** the rejection region for the chosen α
6. **Compute** the test statistic value from sample data
7. **Decide** whether to reject H_0 and state the conclusion in context

Seven-Step Procedure (Devore 9.2)

Recommended Steps for Any Hypothesis Test

1. **Identify** the parameter of interest and describe it in context
2. **State** the null and alternative hypotheses
3. **Check** assumptions for the test procedure
4. **Write** the formula for the test statistic
5. **State** the rejection region for the chosen α
6. **Compute** the test statistic value from sample data
7. **Decide** whether to reject H_0 and state the conclusion in context

We will use this structured approach throughout the course.

Example: Neonatal Birth Weight Screening

Setting: A neonatal unit monitors the average birth weight of infants. Historically, the mean birth weight in this hospital is 3300 g with known $\sigma = 450$ g. After a change in prenatal care protocol, a sample of $n = 36$ births yields $\bar{x} = 3180$ g.

Question: Is there evidence that the new protocol has changed the mean birth weight?

Example: Neonatal Birth Weight Screening

Setting: A neonatal unit monitors the average birth weight of infants. Historically, the mean birth weight in this hospital is 3300 g with known $\sigma = 450$ g. After a change in prenatal care protocol, a sample of $n = 36$ births yields $\bar{x} = 3180$ g.

Question: Is there evidence that the new protocol has changed the mean birth weight?

Step 1: μ = true mean birth weight (g) under the new protocol

Step 2: $H_0 : \mu = 3300$ vs $H_a : \mu \neq 3300$ (two-tailed – either direction is concerning)

Step 3: $\sigma = 450$ is known; $n = 36$ is large enough for CLT

Example: Neonatal Birth Weight (continued)

Step 4: Test statistic:

$$Z = \frac{\bar{X} - 3300}{450/\sqrt{36}} = \frac{\bar{X} - 3300}{75}$$

Step 5: For $\alpha = 0.05$: Reject H_0 if $|z| \geq z_{0.025} = 1.96$

```
1 # Step 6: Compute test statistic
2 xbar <- 3180
3 mu0 <- 3300
4 sigma <- 450
5 n <- 36
6 se <- sigma / sqrt(n)
7
8 z_stat <- (xbar - mu0) / se
9 cat("Test statistic: z =", round(z_stat, 3), "\n")
```

Test statistic: z = -1.6

```
1 # Critical value
2 z_crit <- qnorm(1 - 0.05/2)
3 cat("Critical value: z_{alpha/2} =", round(z_crit, 3), "\n")
```

Critical value: $z_{\alpha/2} = 1.96$



Example: Neonatal Birth Weight (decision)

```
1 # Step 7: Decision
2 cat("|z| =", round(abs(z_stat), 3), "\n")
```

|z| = 1.6

```
1 cat("Critical value =", round(z_crit, 3), "\n\n")
```

Critical value = 1.96

```
1 if (abs(z_stat) >= z_crit) {
2   cat("Decision: Reject H0 at alpha = 0.05\n\n")
3   cat("Conclusion: There is statistically significant evidence that\n")
4   cat("the true mean birth weight under the new protocol differs from 3300 g.")
5 } else {
6   cat("Decision: Fail to reject H0 at alpha = 0.05\n\n")
7   cat("Conclusion: There is insufficient evidence that the true mean\n")
8   cat("birth weight differs from 3300 g under the new protocol.")
9 }
```

Decision: Fail to reject H_0 at $\alpha = 0.05$

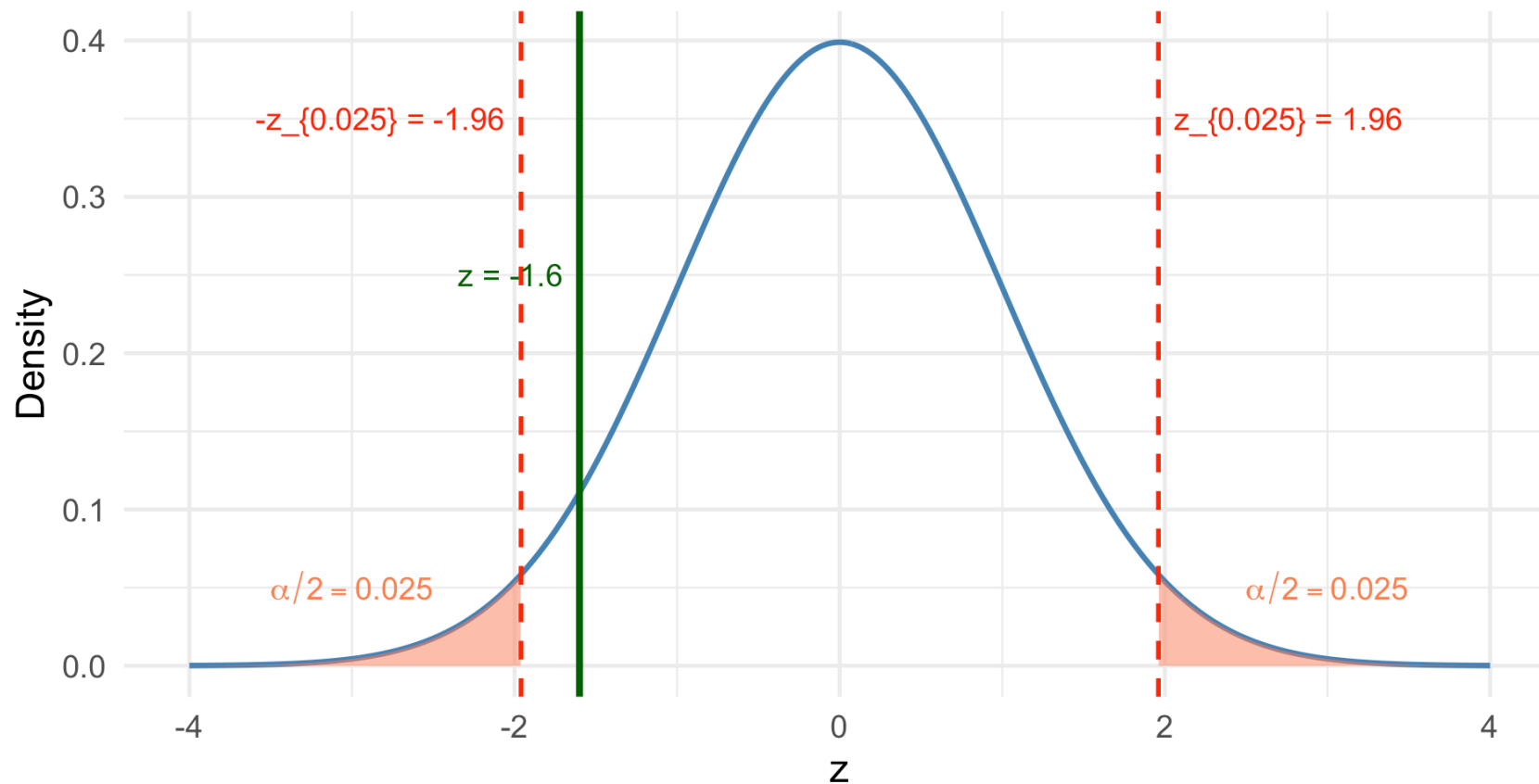
Conclusion: There is insufficient evidence that the true mean birth weight differs from 3300 g under the new protocol.

Visualizing the Test

► Code

Two-Tailed z Test: Neonatal Birth Weight

Test statistic falls in rejection region $\rightarrow$ Reject H_0



Confirming with a Confidence Interval

Recall the **CI-test duality**: the 95% CI contains all μ_0 values that would not be rejected.

```
1 # 95% CI for mu
2 ci_lower <- xbar - z_crit * se
3 ci_upper <- xbar + z_crit * se
4 cat("95% CI for mu: (", round(ci_lower, 1), ",", round(ci_upper, 1), ")\n")
```

95% CI for mu: (3033 , 3327)

```
1 cat("Null value mu_0 = 3300\n\n")
```

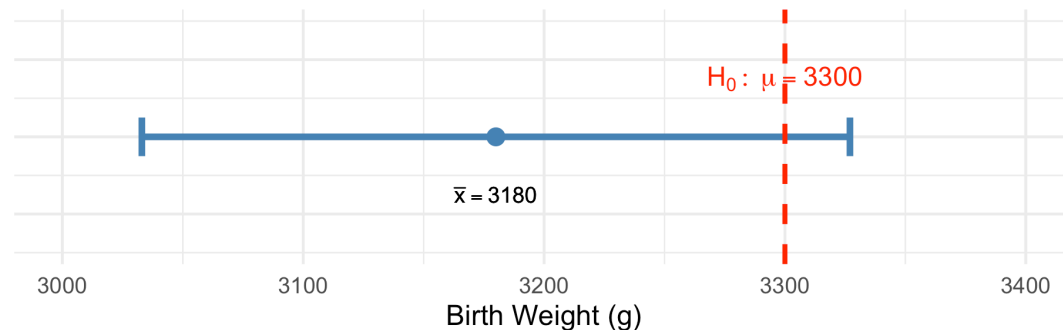
Null value $\mu_0 = 3300$

```
1 cat("Since 3300 is NOT in the CI, we reject H0 -- consistent with our test!")
```

Since 3300 is NOT in the CI, we reject H_0 -- consistent with our test!

► Code

95% Confidence Interval vs Null Value



Type II Error and Power for the z Test (Devore 9.2)

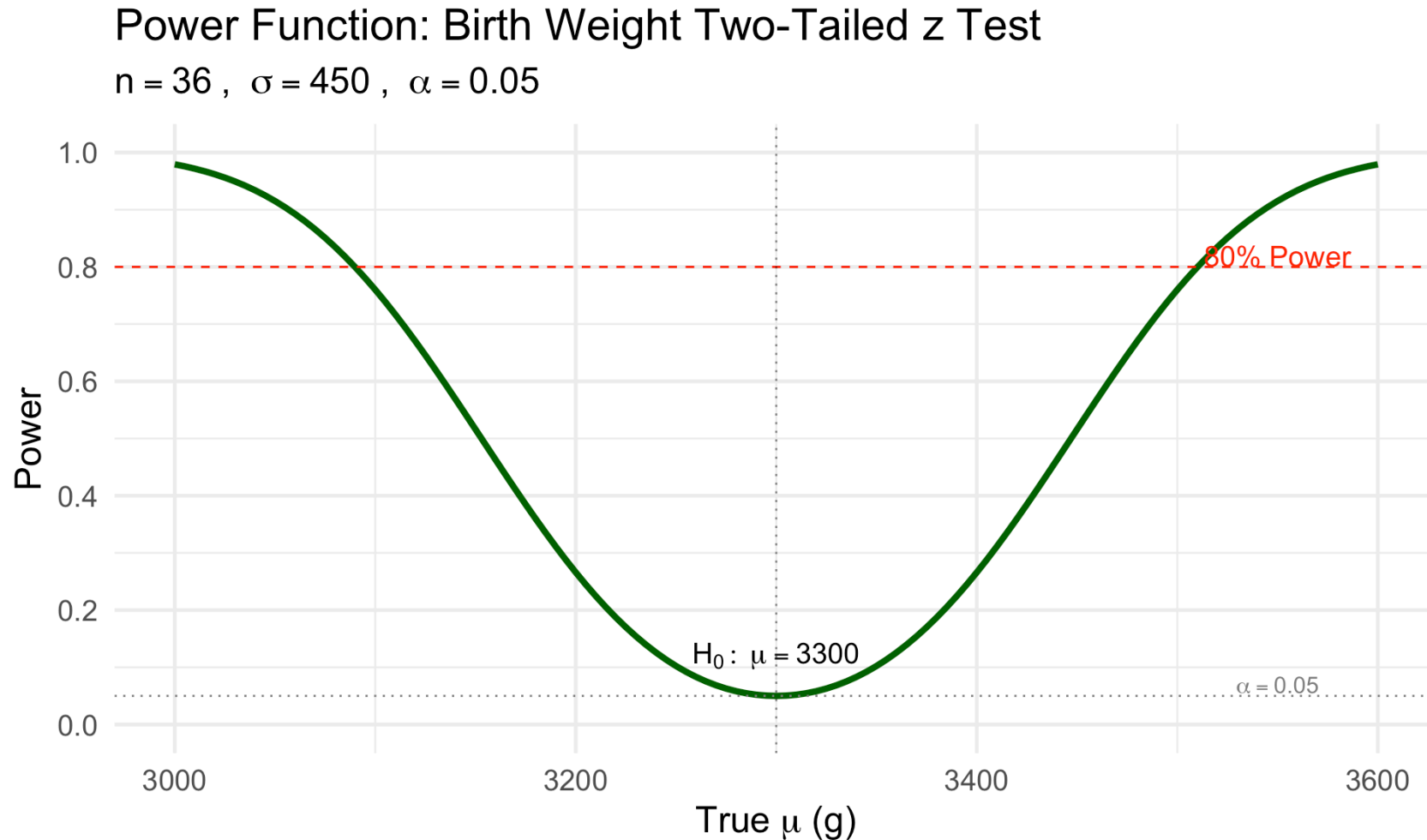
For the z test, we have closed-form expressions for $\beta(\mu')$:

Alternative	$\beta(\mu')$
$H_a : \mu > \mu_0$	$\Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$
$H_a : \mu < \mu_0$	$1 - \Phi\left(-z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$
$H_a : \mu \neq \mu_0$	$\Phi\left(z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) - \Phi\left(-z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$

where $\Phi(z)$ is the standard normal CDF. Power = $1 - \beta(\mu')$.

Power Curve: Birth Weight Example

► Code



Sample Size Determination for the z Test

The sample size n for which a level α test also satisfies $\beta(\mu') = \beta$ is:

$$n = \begin{cases} \left[\frac{\sigma(z_\alpha + z_\beta)}{\mu_0 - \mu'} \right]^2 & \text{one-tailed test} \\ \left[\frac{\sigma(z_{\alpha/2} + z_\beta)}{\mu_0 - \mu'} \right]^2 & \text{two-tailed test (approx)} \end{cases}$$

```
1 # How many births to detect a 100g decrease with 80% power?
2 delta <- 100 # clinically meaningful difference
3 alpha <- 0.05
4 beta_target <- 0.20 # want power = 0.80
5 sigma <- 450
6
7 # Two-tailed test
8 n_needed <- (sigma * (qnorm(1 - alpha/2) + qnorm(1 - beta_target)) / delta)^2
9 cat("Sample size needed:", ceiling(n_needed), "births")
```

Sample size needed: 159 births

Sample Size Determination for the z Test

The sample size n for which a level α test also satisfies $\beta(\mu') = \beta$ is:

$$n = \begin{cases} \left[\frac{\sigma(z_\alpha + z_\beta)}{\mu_0 - \mu'} \right]^2 & \text{one-tailed test} \\ \left[\frac{\sigma(z_{\alpha/2} + z_\beta)}{\mu_0 - \mu'} \right]^2 & \text{two-tailed test (approx)} \end{cases}$$

```
1 # How many births to detect a 100g decrease with 80% power?
2 delta <- 100 # clinically meaningful difference
3 alpha <- 0.05
4 beta_target <- 0.20 # want power = 0.80
5 sigma <- 450
6
7 # Two-tailed test
8 n_needed <- (sigma * (qnorm(1 - alpha/2) + qnorm(1 - beta_target)) / delta)^2
9 cat("Sample size needed:", ceiling(n_needed), "births")
```

Sample size needed: 159 births

Clinical Relevance

In medical research, we should determine the **clinically meaningful difference** before planning the study. A 100 g difference in birth weight may be clinically important; a 10 g difference likely is not.



The One-Sample t Test (Devore 9.2)

From z to t: The Realistic Case

The Problem: In practice, we rarely know σ .

From z to t: The Realistic Case

The Problem: In practice, we rarely know σ .

Solution: Replace σ with the sample standard deviation S :

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \longrightarrow T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

From z to t: The Realistic Case

The Problem: In practice, we **rarely know** σ .

Solution: Replace σ with the sample standard deviation S :

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \longrightarrow T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

Key result (from Lesson 8): When sampling from a normal population,

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad \text{when } H_0 \text{ is true}$$

The t_{n-1} distribution has **heavier tails** than the standard normal, reflecting the additional uncertainty from estimating σ .

The One-Sample t Test

⚠ One-Sample t Test (Devore 9.2)

Null hypothesis: $H_0 : \mu = \mu_0$

Test statistic: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Alternative	Rejection Region
$H_a : \mu > \mu_0$	$t \geq t_{\alpha,n-1}$
$H_a : \mu < \mu_0$	$t \leq -t_{\alpha,n-1}$
$H_a : \mu \neq \mu_0$	$ t \geq t_{\alpha/2,n-1}$

The One-Sample t Test

! One-Sample t Test (Devore 9.2)

Null hypothesis: $H_0 : \mu = \mu_0$

Test statistic: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Alternative	Rejection Region
$H_a : \mu > \mu_0$	$t \geq t_{\alpha, n-1}$
$H_a : \mu < \mu_0$	$t \leq -t_{\alpha, n-1}$
$H_a : \mu \neq \mu_0$	$ t \geq t_{\alpha/2, n-1}$

Assumptions:

- The data are a random sample from a normal distribution
- For large n (say $n > 40$), the test is robust to non-normality

z vs t: Comparison of Critical Values

► Code

```
# A tibble: 6 × 3
  df `t critical value` `z critical value`
  <dbl>                <dbl>          <dbl>
1     5                2.57            1.96
2    10                2.23            1.96
3    20                2.09            1.96
4    50                2.01            1.96
5   100                1.98            1.96
6   Inf                1.96            1.96
```

z vs t: Comparison of Critical Values

► Code

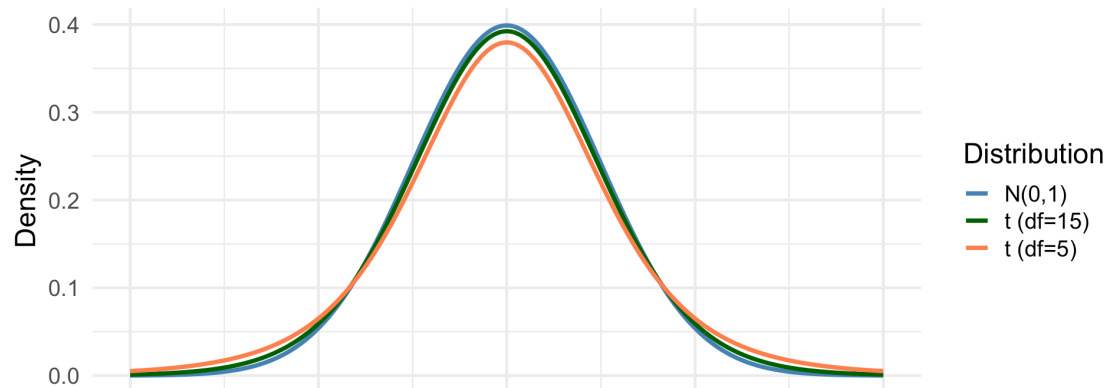
```
# A tibble: 6 × 3
```

	df	`t critical value`	`z critical value`
	<dbl>	<dbl>	<dbl>
1	5	2.57	1.96
2	10	2.23	1.96
3	20	2.09	1.96
4	50	2.01	1.96
5	100	1.98	1.96
6	Inf	1.96	1.96

► Code

Standard Normal vs t Distributions

t distributions have heavier tails, especially with small df



Example: Hemoglobin A1c Monitoring (Devore 9.2 Style)

Setting: A diabetes clinic monitors glycated hemoglobin (HbA1c). The target for well-controlled diabetes is $\text{HbA1c} \leq 7.0\%$. A sample of $n = 15$ patients on a new medication yields:

```
1 hba1c <- c(6.8, 7.2, 6.5, 7.8, 6.9, 7.4, 6.3, 7.1, 6.7,  
2           7.5, 6.6, 7.0, 7.3, 6.4, 7.6)  
3  
4 # Summary statistics  
5 xbar <- mean(hba1c)  
6 s <- sd(hba1c)  
7 n <- length(hba1c)  
8 cat("n =", n, "\n")
```

$n = 15$

```
1 cat("x-bar =", round(xbar, 3), "\n")
```

$\bar{x} = 7.007$

```
1 cat("s =", round(s, 3))
```

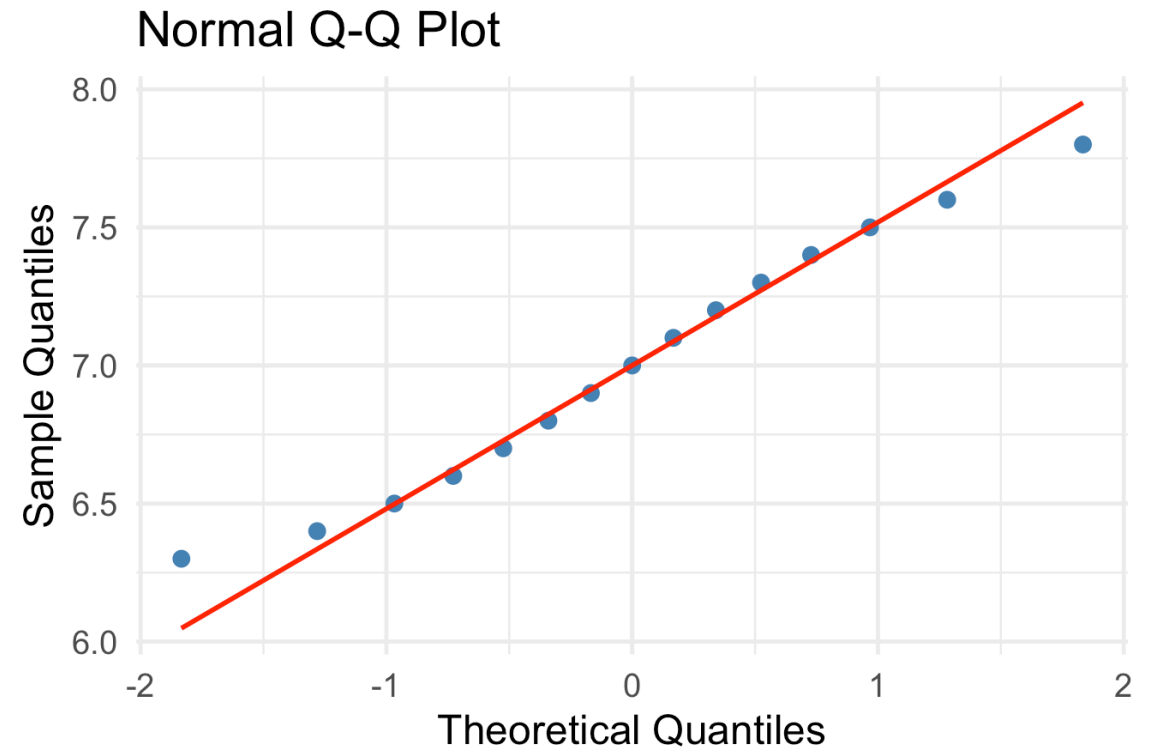
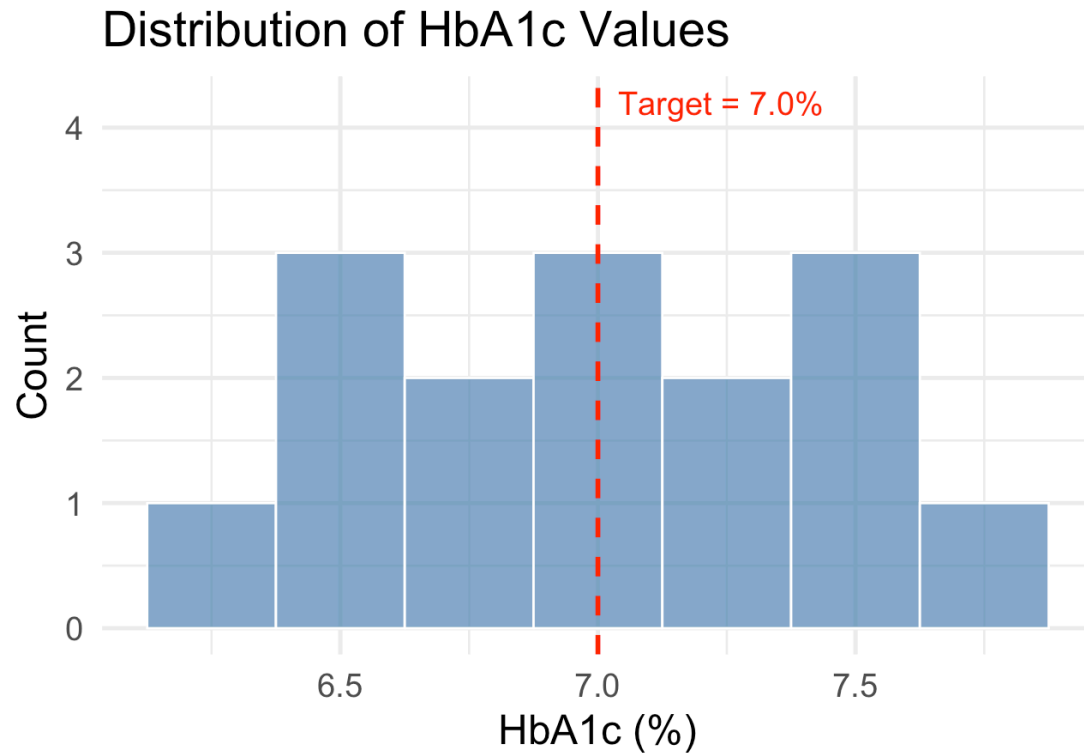
$s = 0.459$

Question: Is there evidence that the true mean HbA1c for patients on this medication exceeds the 7.0% target?

HbA1c Example: Check Assumptions

Step 3: Check normality assumption.

► Code



The Q-Q plot shows approximately linear pattern — normality assumption is reasonable for $n = 15$.

HbA1c Example: Conduct the t Test

```
1 # Step 1-2: mu = true mean HbA1c; H0: mu = 7.0 vs Ha: mu > 7.0
2 mu0 <- 7.0
3 alpha <- 0.05
4
5 # Step 4: Test statistic
6 se <- s / sqrt(n)
7 t_stat <- (xbar - mu0) / se
8 cat("Test statistic: t =", round(t_stat, 3), "\n")
```

Test statistic: t = 0.056

```
1 # Step 5: Critical value (upper-tailed, df = n-1)
2 t_crit <- qt(1 - alpha, df = n - 1)
3 cat("Critical value: t_{0.05, 14} =", round(t_crit, 3), "\n")
```

Critical value: $t_{\{0.05, 14\}} = 1.761$

```
1 # Step 7: Decision
2 cat("\nDecision:",
3     ifelse(t_stat >= t_crit, "Reject H0", "Fail to reject H0"))
```

Decision: Fail to reject H_0

HbA1c Example: Conduct the t Test

```
1 # Step 1-2: mu = true mean HbA1c; H0: mu = 7.0 vs Ha: mu > 7.0
2 mu0 <- 7.0
3 alpha <- 0.05
4
5 # Step 4: Test statistic
6 se <- s / sqrt(n)
7 t_stat <- (xbar - mu0) / se
8 cat("Test statistic: t =", round(t_stat, 3), "\n")
```

Test statistic: t = 0.056

```
1 # Step 5: Critical value (upper-tailed, df = n-1)
2 t_crit <- qt(1 - alpha, df = n - 1)
3 cat("Critical value: t_{0.05, 14} =", round(t_crit, 3), "\n")
```

Critical value: t_{0.05, 14} = 1.761

```
1 # Step 7: Decision
2 cat("\nDecision:",
3     ifelse(t_stat >= t_crit, "Reject H0", "Fail to reject H0"))
```

Decision: Fail to reject H0

Step 7 Interpretation: There is insufficient evidence at $\alpha = 0.05$ to conclude that the true mean HbA1c exceeds the 7.0% target.

Using R's `t.test()` Function

```
1 # R's built-in t.test function
2 t_result <- t.test(hba1c, mu = 7.0, alternative = "greater")
3 t_result
```

One Sample t-test

```
data:  hba1c
t = 0.056254, df = 14, p-value = 0.478
alternative hypothesis: true mean is greater than 7
95 percent confidence interval:
 6.797935      Inf
sample estimates:
mean of x
 7.006667
```


Using R's `t.test()` Function

```
1 # R's built-in t.test function
2 t_result <- t.test(hba1c, mu = 7.0, alternative = "greater")
3 t_result
```

One Sample t-test

```
data: hba1c
t = 0.056254, df = 14, p-value = 0.478
alternative hypothesis: true mean is greater than 7
95 percent confidence interval:
 6.797935      Inf
sample estimates:
mean of x
 7.006667
```

```
1 # Extract tidy results
2 tidy(t_result)
```

A tibble: 1 × 8

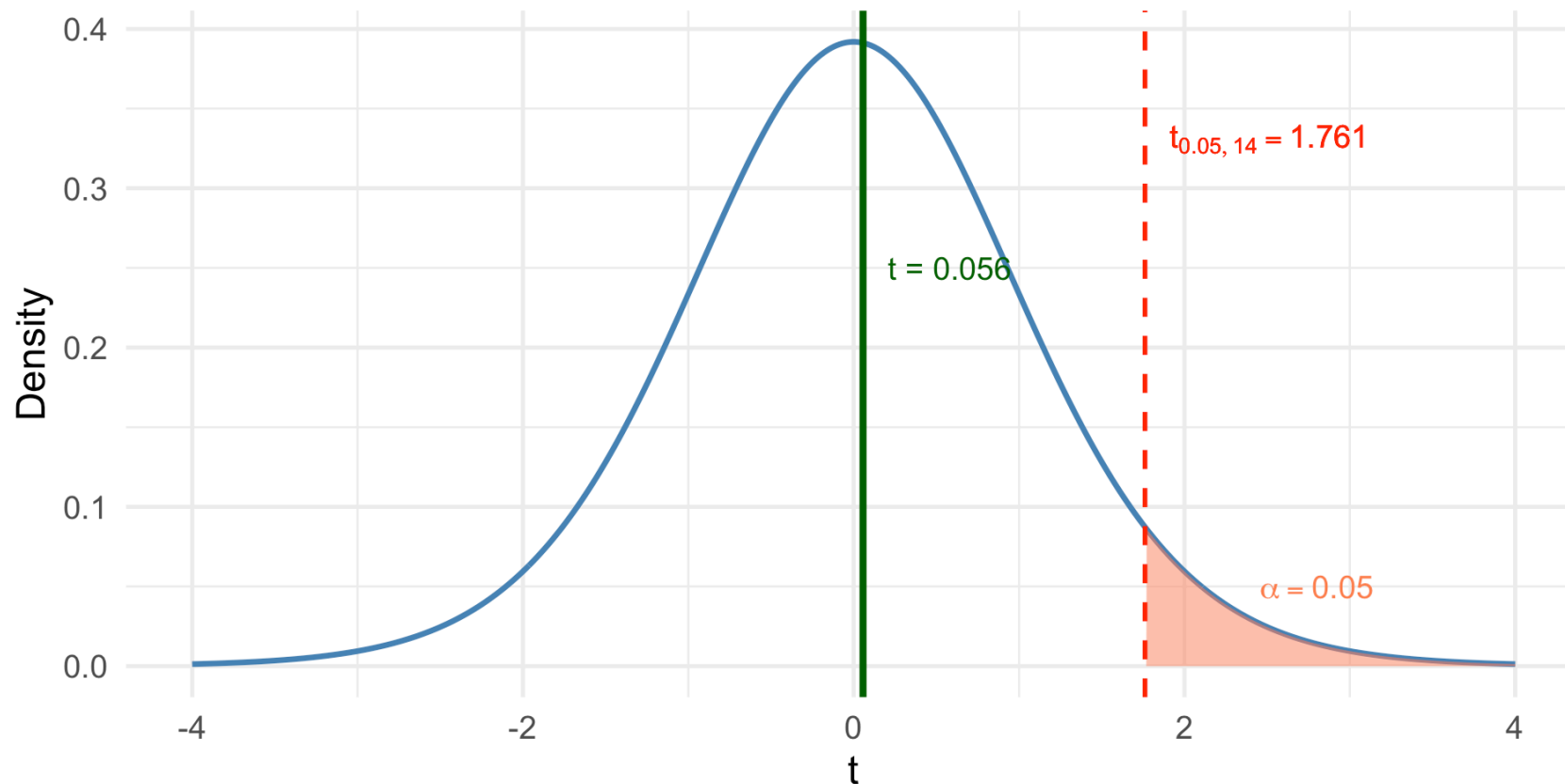
	estimate	statistic	p.value	parameter	conf.low	conf.high	method	alternative
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1	7.01	0.0563	0.478	14	6.80	Inf	One Sampl...	greater

Visualizing the t Test Result

► Code

Upper-Tailed t Test: HbA1c (t_{14} distribution)

Test statistic does NOT fall in rejection region → Fail to reject H_0



Example: Body Temperature Revisited (Two-Tailed t Test)

The widely cited “normal” body temperature of 98.6°F comes from a 19th century study. Mackowiak et al. (1992, JAMA) measured temperatures in 130 healthy adults.

```
1 # Simulated data consistent with the Mackowiak et al. findings
2 set.seed(1992)
3 body_temp <- rnorm(130, mean = 98.2, sd = 0.73)
4
5 cat("n =", length(body_temp), "\n")
```

n = 130

```
1 cat("x-bar =", round(mean(body_temp), 3), "\n")
```

x-bar = 98.195

```
1 cat("s =", round(sd(body_temp), 3))
```

s = 0.668

Example: Body Temperature Revisited (Two-Tailed t Test)

The widely cited “normal” body temperature of 98.6°F comes from a 19th century study. Mackowiak et al. (1992, JAMA) measured temperatures in 130 healthy adults.

```
1 # Simulated data consistent with the Mackowiak et al. findings
2 set.seed(1992)
3 body_temp <- rnorm(130, mean = 98.2, sd = 0.73)
4
5 cat("n =", length(body_temp), "\n")
```

n = 130

```
1 cat("x-bar =", round(mean(body_temp), 3), "\n")
```

x-bar = 98.195

```
1 cat("s =", round(sd(body_temp), 3))
```

s = 0.668

Hypotheses: $H_0 : \mu = 98.6$ vs $H_a : \mu \neq 98.6$

Body Temperature: t Test in R

```
1 # Two-tailed t test
2 temp_test <- t.test(body_temp, mu = 98.6, alternative = "two.sided")
3 temp_test
```

One Sample t-test

```
data:  body_temp
t = -6.9161, df = 129, p-value = 1.934e-10
alternative hypothesis: true mean is not equal to 98.6
95 percent confidence interval:
 98.07896 98.31076
sample estimates:
mean of x
 98.19486
```

Body Temperature: t Test in R

```
1 # Two-tailed t test
2 temp_test <- t.test(body_temp, mu = 98.6, alternative = "two.sided")
3 temp_test
```

One Sample t-test

```
data: body_temp
t = -6.9161, df = 129, p-value = 1.934e-10
alternative hypothesis: true mean is not equal to 98.6
95 percent confidence interval:
 98.07896 98.31076
sample estimates:
mean of x
 98.19486
```

```
1 # Extract key quantities
2 cat("t statistic:", round(temp_test$statistic, 3), "\n")
```

t statistic: -6.916

```
1 cat("p-value:", format.pval(temp_test$p.value, digits = 3), "\n")
```

p-value: 1.93e-10

```
1 cat("95% CI: (", round(temp_test$conf.int[1], 3), ",",
2     round(temp_test$conf.int[2], 3), ")\n\n")
```

When Can We Use the t Test?

! Guidelines for the One-Sample t Test	
Normality is critical when n is small:	
Sample Size	Guideline
$n < 15$	Use t test only if data appear close to normal (check Q-Q plot)
$15 \leq n \leq 40$	Mild skewness is acceptable; avoid if heavily skewed or outliers present
$n > 40$	t test is robust to non-normality by CLT; can use for most distributions

When Can We Use the t Test?

 Guidelines for the One-Sample t Test

Normality is critical when n is small:

Sample Size	Guideline
$n < 15$	Use t test only if data appear close to normal (check Q-Q plot)
$15 \leq n \leq 40$	Mild skewness is acceptable; avoid if heavily skewed or outliers present
$n > 40$	t test is robust to non-normality by CLT; can use for most distributions

 Practical Advice (Devore 9.2)

For moderate sample sizes, examine a **normal probability plot** (Q-Q plot) before using the t test. If the data show substantial departures from normality, consider a **bootstrap test** (Lesson 15) or a **nonparametric test** instead.

Simulation: Coverage of the t Test Under Non-Normality

► Code

```
# A tibble: 6 × 3
  Distribution      n `Type I Error Rate`
  <chr>          <dbl>          <dbl>
1 Normal         10            1
2 Normal         50            1
3 Exp (shifted)   10         0.0989
4 Exp (shifted)   50         0.066
5 t(3) (shifted)  10         0.0394
6 t(3) (shifted)  50         0.047
```

Simulation: Coverage of the t Test Under Non-Normality

► Code

```
# A tibble: 6 × 3
  Distribution      n `Type I Error Rate`
  <chr>          <dbl>          <dbl>
1 Normal         10            1
2 Normal         50            1
3 Exp (shifted)   10          0.0989
4 Exp (shifted)   50          0.066
5 t(3) (shifted)  10          0.0394
6 t(3) (shifted)  50          0.047
```

The t test maintains $\alpha \approx 0.05$ well for normal and heavy-tailed symmetric distributions, but may be anticonservative for skewed distributions with small n .

Textbook Example: Particle Size (Devore Example 9.12)

Devore Example 9.12: Particulate Matter

Roadside measurements of particle sizes (d_{50} in microns) at $n = 9$ assays near a North Carolina highway:

```
1 d50 <- c(82.9, 56.8, 66.5, 49.4, 105.4, 79.5, 82.5, 50.7, 43.0)
2
3 # Summary statistics
4 cat("n =", length(d50), "\n")
```

$n = 9$

```
1 cat("x-bar =", round(mean(d50), 2), "\n")
```

$\bar{x} = 68.52$

```
1 cat("s =", round(sd(d50), 2))
```

$s = 20.49$

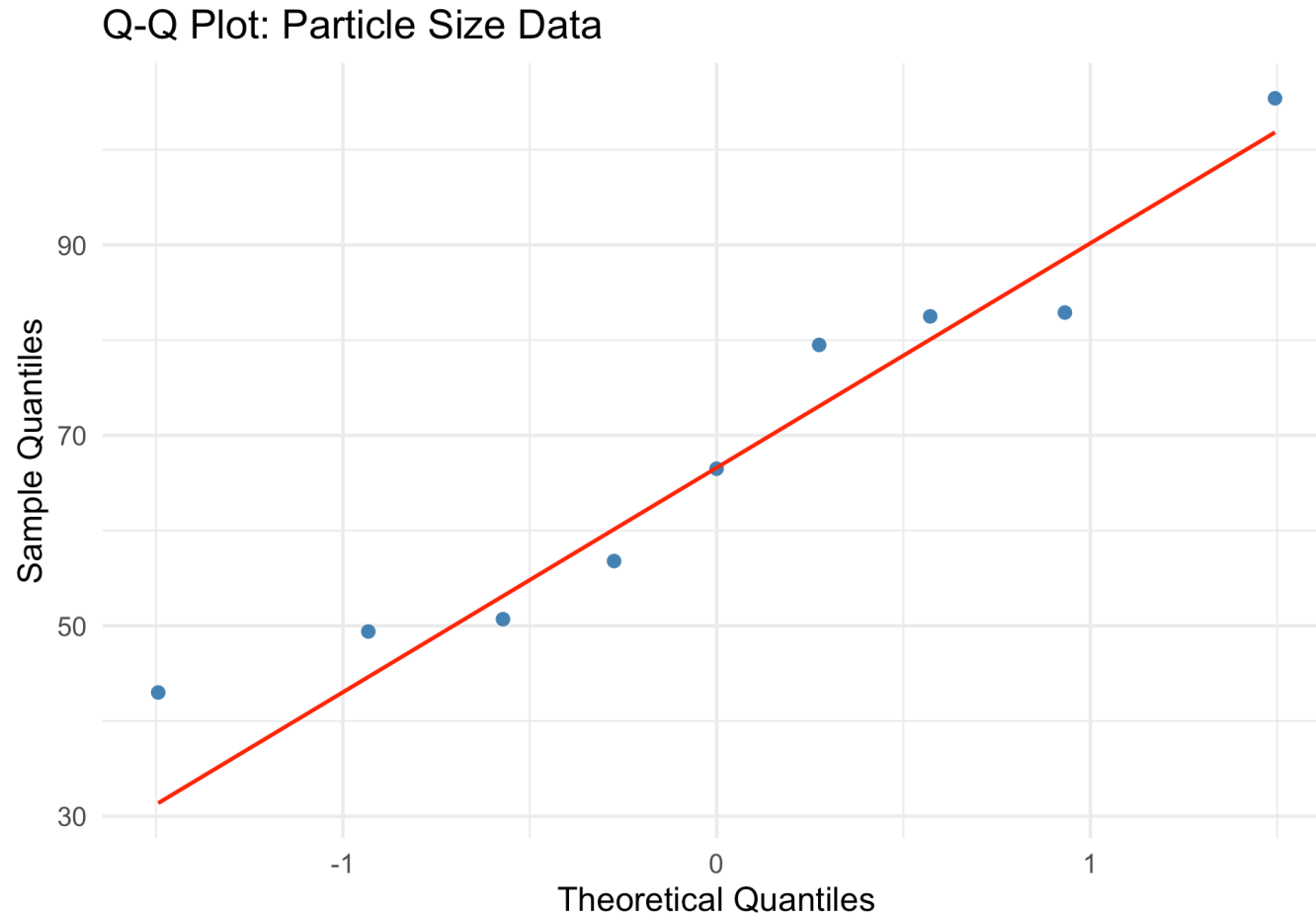
Previous studies indicated that the typical d_{50} value alongside similar roads is 44 microns.

Hypotheses: $H_0 : \mu = 44$ vs $H_a : \mu \neq 44$ at $\alpha = 0.01$.

Devore Example 9.12: Solution

1 # Check normality

► Code



Devore Example 9.12: Solution

```
1 # t test
2 t.test(d50, mu = 44, alternative = "two.sided", conf.level = 0.99)
```

One Sample t-test

```
data: d50
t = 3.5904, df = 8, p-value = 0.007081
alternative hypothesis: true mean is not equal to 44
99 percent confidence interval:
 45.60481 91.43964
sample estimates:
mean of x
 68.52222
```

Devore Example 9.12: Solution

```
1 # t test
2 t.test(d50, mu = 44, alternative = "two.sided", conf.level = 0.99)
```

One Sample t-test

```
data: d50
t = 3.5904, df = 8, p-value = 0.007081
alternative hypothesis: true mean is not equal to 44
99 percent confidence interval:
 45.60481 91.43964
sample estimates:
mean of x
 68.52222
```

Since $|t| = 3.59 > t_{0.005,8} = 3.355$, we reject H_0 at $\alpha = 0.01$. The data provides strong evidence that the true mean d_{50} at this site differs from 44 microns.

Textbook Example: Tip Percentages (Devore Example 9.13)

Devore Example 9.13: Large-Sample t Test

A sample of $n = 70$ tip percentages at a restaurant. Summary statistics: $\bar{x} = 17.986$, $s = 5.937$.

Question: Is there evidence that the true average tip percentage exceeds 15%?

```
1 # Given summary statistics (large n, don't need raw data)
2 xbar_tip <- 17.986
3 s_tip <- 5.937
4 n_tip <- 70
5 mu0_tip <- 15
6 alpha <- 0.05
7
8 # t test statistic
9 se_tip <- s_tip / sqrt(n_tip)
10 t_stat_tip <- (xbar_tip - mu0_tip) / se_tip
11 t_crit_tip <- qt(1 - alpha, df = n_tip - 1)
12
13 cat("t =", round(t_stat_tip, 2), "\n")
```

t = 4.21

```
1 cat("t_{0.05, 69} =", round(t_crit_tip, 3), "\n")
```

t_{0.05, 69} = 1.667

```
1 cat("Decision:", ifelse(t_stat_tip >= t_crit_tip, "Reject H0", "Fail to reject H0"))
```

Decision: Reject H0



Devore Example 9.13: Large-Sample t Test

A sample of $n = 70$ tip percentages at a restaurant. Summary statistics: $\bar{x} = 17.986$, $s = 5.937$.

Question: Is there evidence that the true average tip percentage exceeds 15%?

```
1 # Given summary statistics (large n, don't need raw data)
2 xbar_tip <- 17.986
3 s_tip <- 5.937
4 n_tip <- 70
5 mu0_tip <- 15
6 alpha <- 0.05
7
8 # t test statistic
9 se_tip <- s_tip / sqrt(n_tip)
10 t_stat_tip <- (xbar_tip - mu0_tip) / se_tip
11 t_crit_tip <- qt(1 - alpha, df = n_tip - 1)
12
13 cat("t =", round(t_stat_tip, 2), "\n")
```

t = 4.21

```
1 cat("t_{0.05, 69} =", round(t_crit_tip, 3), "\n")
```

t_{0.05, 69} = 1.667

```
1 cat("Decision:", ifelse(t_stat_tip >= t_crit_tip, "Reject H0", "Fail to reject H0"))
```

Decision: Reject H0



Tests About a Population Proportion (Devore 9.3)

Testing a Population Proportion

Setting: We want to test a claim about the population proportion p of “successes.”

Data: X = number of successes in a random sample of size n , where $X \sim \text{Binomial}(n, p)$.

Testing a Population Proportion

Setting: We want to test a claim about the population proportion p of “successes.”

Data: X = number of successes in a random sample of size n , where $X \sim \text{Binomial}(n, p)$.

Estimator: $\hat{p} = X/n$ (sample proportion)

Testing a Population Proportion

Setting: We want to test a claim about the population proportion p of “successes.”

Data: X = number of successes in a random sample of size n , where $X \sim \text{Binomial}(n, p)$.

Estimator: $\hat{p} = X/n$ (sample proportion)

When $H_0 : p = p_0$ is true and n is large:

$$\hat{P} \approx N\left(p_0, \frac{p_0(1 - p_0)}{n}\right)$$

Standardizing gives the test statistic:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

The One-Proportion z Test

! One-Proportion z Test (Devore 9.3)

Null hypothesis: $H_0 : p = p_0$

Test statistic: $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$

Alternative	Rejection Region
$H_a : p > p_0$	$z \geq z_\alpha$
$H_a : p < p_0$	$z \leq -z_\alpha$
$H_a : p \neq p_0$	$ z \geq z_{\alpha/2}$

Validity condition: $np_0 \geq 10$ and $n(1 - p_0) \geq 10$

The One-Proportion z Test

! One-Proportion z Test (Devore 9.3)

Null hypothesis: $H_0 : p = p_0$

Test statistic: $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$

Alternative	Rejection Region
$H_a : p > p_0$	$z \geq z_\alpha$
$H_a : p < p_0$	$z \leq -z_\alpha$
$H_a : p \neq p_0$	$ z \geq z_{\alpha/2}$

Validity condition: $np_0 \geq 10$ and $n(1 - p_0) \geq 10$

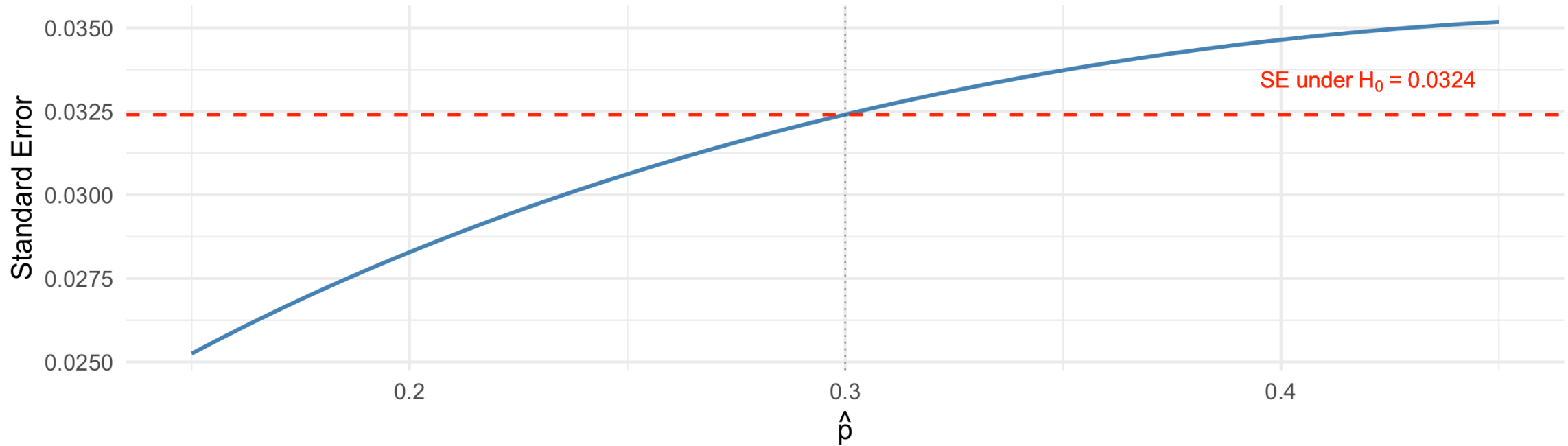
Key point: Under H_0 , we use p_0 (not $\hat{p}$) in the standard error! This differs from the CI formula where we used $\hat{p}$ (Wald) or the score approach.

Why p_0 in the Standard Error?

► Code

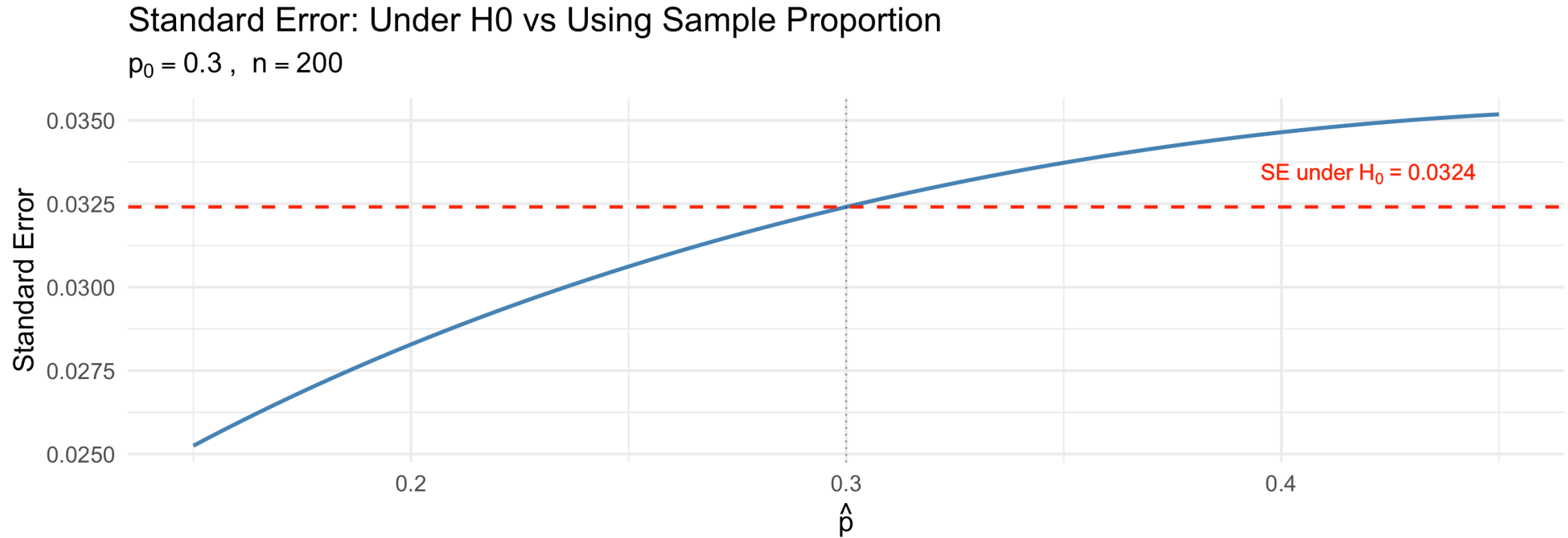
Standard Error: Under H_0 vs Using Sample Proportion

$p_0 = 0.3$, $n = 200$



Why p_0 in the Standard Error?

► Code



Reasoning: When **testing**, we assume H_0 is true and ask “how unlikely is the data?” So we use p_0 . When **estimating** (CI), we don’t assume a specific value for p , so we use $\hat{p}$.

Example: Post-Surgical Infection Rate

Setting: The national rate of post-surgical site infections is 2.5% for a certain procedure. A hospital implements a new sterilization protocol and observes 8 infections among 500 surgeries.

Question: Has the new protocol reduced the infection rate?

Example: Post-Surgical Infection Rate

Setting: The national rate of post-surgical site infections is 2.5% for a certain procedure. A hospital implements a new sterilization protocol and observes 8 infections among 500 surgeries.

Question: Has the new protocol reduced the infection rate?

```
1 # Data
2 x <- 8           # infections observed
3 n <- 500         # total surgeries
4 p_hat <- x / n
5 p0 <- 0.025      # national rate
6
7 cat("Sample proportion: p-hat =", p_hat, "\n")
```

Sample proportion: p-hat = 0.016

```
1 cat("Expected under H0:", n * p0, "infections\n")
```

Expected under H0: 12.5 infections

```
1 # Check validity: np0 and n(1-p0)
2 cat("np0 =", n * p0, ">= 10?", n * p0 >= 10, "\n")
```

np0 = 12.5 >= 10? TRUE

```
1 cat("n(1-p0) =", n * (1 - p0), ">= 10?", n * (1 - p0) >= 10)
```

n(1-p0) = 487.5 >= 10? TRUE

Infection Rate: Conduct the Test

```
1 # H0: p = 0.025 vs Ha: p < 0.025 (lower-tailed)
2 alpha <- 0.05
3
4 # Test statistic
5 se_h0 <- sqrt(p0 * (1 - p0) / n)
6 z_stat <- (p_hat - p0) / se_h0
7 z_crit <- qnorm(alpha) # lower-tailed critical value
8
9 cat("z =", round(z_stat, 3), "\n")
```

z = -1.289

```
1 cat("Critical value: -z_{0.05} =", round(z_crit, 3), "\n\n")
```

Critical value: $-z_{0.05} = -1.645$

```
1 if (z_stat <= z_crit) {
2   cat("Decision: Reject H0\n")
3   cat("Conclusion: Evidence that the new protocol reduces infection rate.")
4 } else {
5   cat("Decision: Fail to reject H0\n")
6   cat("Conclusion: Insufficient evidence that the new protocol\n")
7   cat("reduces infection rate below the national 2.5%.")
8 }
```

Decision: Fail to reject H_0

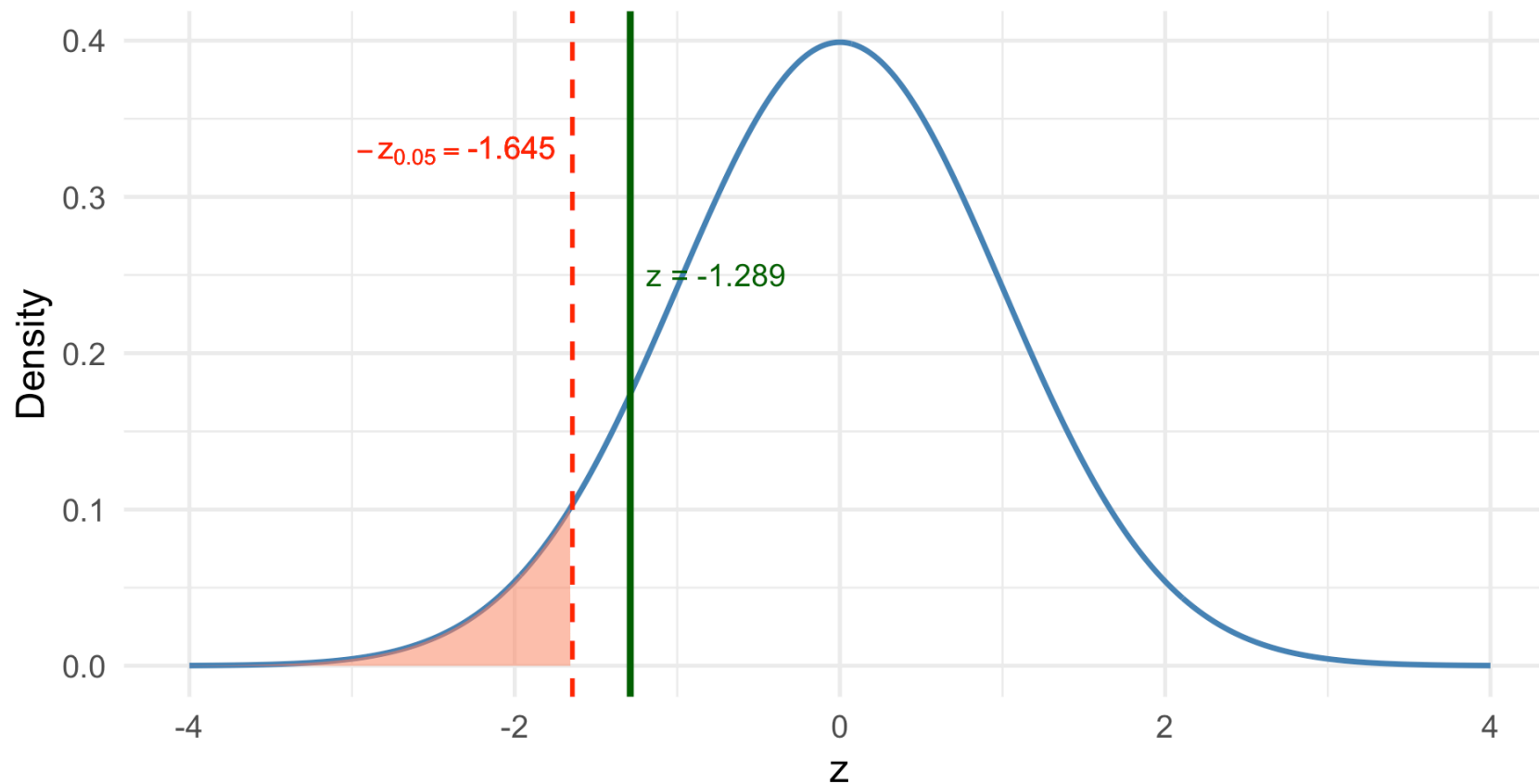


Visualizing the Proportion Test

► Code

Lower-Tailed z Test: Post-Surgical Infection Rate

Test statistic does NOT fall in rejection region → Fail to reject H_0



What Does `prop.test()` Actually Do?

R's `prop.test()` performs the **score test** for a population proportion — a test based on the score pivot used in the score interval in Lesson 8. Recall that the pivot:

$$Z_{score} = \frac{U(\theta)}{\sqrt{I_n(\theta)}} \xrightarrow{d} N(0, 1)$$

What Does `prop.test()` Actually Do?

R's `prop.test()` performs the **score test** for a population proportion — a test based on the score pivot used in the score interval in Lesson 8. Recall that the pivot:

$$Z_{score} = \frac{U(\theta)}{\sqrt{I_n(\theta)}} \xrightarrow{d} N(0, 1)$$

and for a Bernoulli sample:

$$Z_{score} = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

This is exactly the z statistic we computed by hand above. The function `prop.test()` reports the **squared** version under H_0 :

$$X^2 = Z_{score}^2 = \frac{(\hat{p} - p_0)^2}{p_0(1-p_0)/n} \sim \chi_1^2$$

What Does `prop.test()` Actually Do?

R's `prop.test()` performs the **score test** for a population proportion — a test based on the score pivot used in the score interval in Lesson 8. Recall that the pivot:

$$Z_{score} = \frac{U(\theta)}{\sqrt{I_n(\theta)}} \xrightarrow{d} N(0, 1)$$

and for a Bernoulli sample:

$$Z_{score} = \frac{\hat{p} - p}{\sqrt{p(1 - p)/n}}$$

This is exactly the z statistic we computed by hand above. The function `prop.test()` reports the **squared** version under H_0 :

$$X^2 = Z_{score}^2 = \frac{(\hat{p} - p_0)^2}{p_0(1 - p_0)/n} \sim \chi_1^2$$

! Why χ_1^2 ?

If $Z \sim N(0, 1)$, then $Z^2 \sim \chi_1^2$ (chi-squared with 1 df). The score test for proportions can equivalently be expressed as a z test or a χ^2 test — they give exactly

prop.test() is the Score Test

```
1 # Our z test by hand
2 z_stat_manual <- (p_hat - p0) / sqrt(p0 * (1 - p0) / n)
3 cat("Manual z statistic:", round(z_stat_manual, 3), "\n")
```

Manual z statistic: -1.289

```
1 cat("Manual z^2:          ", round(z_stat_manual^2, 3), "\n\n")
```

Manual z^2: 1.662

```
1 # prop.test computes the same thing as chi-squared
2 prop_result <- prop.test(x = 8, n = 500, p = 0.025,
3                           alternative = "less", correct = FALSE)
4
5 cat("prop.test chi-squared:", round(prop_result$statistic, 3), "\n")
```

prop.test chi-squared: 1.662

```
1 cat("sqrt(chi-squared):    ", round(sqrt(prop_result$statistic), 3), "\n")
```

sqrt(chi-squared): 1.289

```
1 cat("p-value (prop.test):  ", round(prop_result$p.value, 4))
```

p-value (prop.test): 0.0987

prop.test() is the Score Test

```
1 # Our z test by hand
2 z_stat_manual <- (p_hat - p0) / sqrt(p0 * (1 - p0) / n)
3 cat("Manual z statistic:", round(z_stat_manual, 3), "\n")
```

Manual z statistic: -1.289

```
1 cat("Manual z^2:      ", round(z_stat_manual^2, 3), "\n\n")
```

Manual z^2: 1.662

```
1 # prop.test computes the same thing as chi-squared
2 prop_result <- prop.test(x = 8, n = 500, p = 0.025,
3                           alternative = "less", correct = FALSE)
4
5 cat("prop.test chi-squared:", round(prop_result$statistic, 3), "\n")
```

prop.test chi-squared: 1.662

```
1 cat("sqrt(chi-squared):  ", round(sqrt(prop_result$statistic), 3), "\n")
```

sqrt(chi-squared): 1.289

```
1 cat("p-value (prop.test): ", round(prop_result$p.value, 4))
```

p-value (prop.test): 0.0987

The X^2 statistic from `prop.test()` equals z^2 from our manual calculation!

Connection to the Score Interval (Lesson 8)

In Lesson 8, we derived the **Score (Wilson) confidence interval** by inverting the score test. Recall:

The score interval collects all values p satisfying:

$$\left| \frac{\hat{p} - p}{\sqrt{p(1 - p)/n}} \right| \leq z_{\alpha/2}$$

Connection to the Score Interval (Lesson 8)

In Lesson 8, we derived the **Score (Wilson) confidence interval** by inverting the score test. Recall:

The score interval collects all values p satisfying:

$$\left| \frac{\hat{p} - p}{\sqrt{p(1 - p)/n}} \right| \leq z_{\alpha/2}$$

The key difference from the Wald interval: the **score** approach keeps the parameter p in the standard error, rather than plugging in $\hat{p}$, and when we are “inverting the test statistic” we no longer use the null p_0 but a general p that we need to solve for.

Connection to the Score Interval (Lesson 8 `prop.test`)

The CI returned by `prop.test()` is exactly the Score (Wilson) interval:

```
1 # prop.test returns the Score (Wilson) CI
2 cat("Score CI from prop.test():\n")
```

Score CI from `prop.test()`:

```
1 cat("  (", round(prop_result$conf.int[1], 5), ",",
2      round(prop_result$conf.int[2], 5), ")\n\n")
```

```
( 0 , 0.02817 )
```

```
1 # Verify: this is the Wilson interval from Lesson 8
2 z_alpha2 <- qnorm(0.975)
3 p_hat_val <- 8/500
4 n_val <- 500
5
6 # Wilson formula: (p_hat + z^2/(2n) +/- z*sqrt(p_hat*(1-p_hat)/n + z^2/(4n^2))) / (1 + z^2/n)
7 center <- (p_hat_val + z_alpha2^2 / (2 * n_val)) / (1 + z_alpha2^2 / n_val)
8 margin <- z_alpha2 * sqrt(p_hat_val * (1 - p_hat_val) / n_val + z_alpha2^2 / (4 * n_val^2))
9      (1 + z_alpha2^2 / n_val)
10
11 cat("Wilson formula by hand:\n")
```

Wilson formula by hand:

```
1 cat("  (", round(center - margin, 5), ",", round(center + margin, 5), ")\n")
```

Three Tests, One Framework

The relationship between the Wald test, Score test, and Likelihood Ratio test is a fundamental concept in statistical inference (we'll see the LRT in Lesson 15):

Approach	SE evaluated at	Test Statistic	CI
Wald	MLE $\hat{p}$	$\frac{\hat{p}-p_0}{\sqrt{\hat{p}(1-\hat{p})/n}}$	Wald CI
Score	Null p_0	$\frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$	Wilson/Score CI

Three Tests, One Framework

The relationship between the Wald test, Score test, and Likelihood Ratio test is a fundamental concept in statistical inference (we'll see the LRT in Lesson 15):

Approach	SE evaluated at	Test Statistic	CI
Wald	MLE $\hat{p}$	$\frac{\hat{p}-p_0}{\sqrt{\hat{p}(1-\hat{p})/n}}$	Wald CI
Score	Null p_0	$\frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$	Wilson/Score CI

 Key Insight

When **testing** $H_0 : p = p_0$, we use p_0 in the SE (score test). The z test for proportions from Devore 9.3 is the score test. The CI from inverting this test is the Score (Wilson) interval from Lesson 8 – not the Wald interval.

prop.test() Options

```
1 # Full output from prop.test (using broom for tidy results)
2 prop.test(x = 8, n = 500, p = 0.025,
3           alternative = "less", correct = FALSE) |>
4   tidy() |>
5   select(estimate, statistic, p.value, conf.low, conf.high, method)
```

A tibble: 1 × 6

	estimate	statistic	p.value	conf.low	conf.high	method
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	0.016	1.66	0.0987	0	0.0282	1-sample proportions test witho...

prop.test() Options

```
1 # Full output from prop.test (using broom for tidy results)
2 prop.test(x = 8, n = 500, p = 0.025,
3           alternative = "less", correct = FALSE) |>
4   tidy() |>
5   select(estimate, statistic, p.value, conf.low, conf.high, method)
```

A tibble: 1 × 6

	estimate	statistic	p.value	conf.low	conf.high	method
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	0.016	1.66	0.0987	0	0.0282	1-sample proportions test witho...

Important arguments:

- `correct = FALSE`: Do **not** apply Yates' continuity correction (matches our z test formula)
- `correct = TRUE` (default): Applies the correction $|\hat{p} - p_0| - \frac{1}{2n}$; slightly conservative
- `alternative`: "two.sided" (default), "less", or "greater"
- The `conf.int` returned is always the **Score (Wilson) CI**, regardless of the `alternative`

The Exact Binomial Test

When the Normal Approximation Fails

The large-sample z test relies on:

$$\frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \dot{\sim} N(0, 1)$$

This approximation requires $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

When the Normal Approximation Fails

The large-sample z test relies on:

$$\frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \dot{\sim} N(0, 1)$$

This approximation requires $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

What do we do when these conditions fail? For example:

- Testing if a rare adverse event rate differs from 1% with only 50 patients ($np_0 = 0.5$)
- Evaluating a new surgical technique on 15 patients

When the Normal Approximation Fails

The large-sample z test relies on:

$$\frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \dot{\sim} N(0, 1)$$

This approximation requires $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

What do we do when these conditions fail? For example:

- Testing if a rare adverse event rate differs from 1% with only 50 patients ($np_0 = 0.5$)
- Evaluating a new surgical technique on 15 patients

Answer: Use the **exact binomial test**, which works directly with the binomial distribution – no approximation needed!

Deriving the Exact Test from First Principles

Setup: Let X = number of successes in n independent Bernoulli trials.

Under $H_0 : p = p_0$, we know exactly:

$$X \sim \text{Binomial}(n, p_0)$$

$$P(X = k \mid p_0) = \binom{n}{k} p_0^k (1 - p_0)^{n-k}, \quad k = 0, 1, \dots, n$$

Deriving the Exact Test from First Principles

Setup: Let X = number of successes in n independent Bernoulli trials.

Under $H_0 : p = p_0$, we know exactly:

$$X \sim \text{Binomial}(n, p_0)$$

$$P(X = k \mid p_0) = \binom{n}{k} p_0^k (1 - p_0)^{n-k}, \quad k = 0, 1, \dots, n$$

The test statistic is simply X itself — the count of successes. No standardization needed!

Deriving the Exact Test from First Principles

Setup: Let X = number of successes in n independent Bernoulli trials.

Under $H_0 : p = p_0$, we know exactly:

$$X \sim \text{Binomial}(n, p_0)$$

$$P(X = k \mid p_0) = \binom{n}{k} p_0^k (1 - p_0)^{n-k}, \quad k = 0, 1, \dots, n$$

The test statistic is simply X itself – the count of successes. No standardization needed!

! Why “Exact”?

The z test is *approximate* because it relies on the CLT approximation $\hat{p} \sim N(p, p(1 - p)/n)$. The binomial test is *exact* because it uses the actual, known distribution of X under H_0 – no asymptotics required.

Rejection Regions for the Exact Binomial Test

For the **upper-tailed** test $H_a : p > p_0$, we reject for large X :

The rejection region is $\{c, c + 1, \dots, n\}$ where c is chosen so that:

$$P(X \geq c \mid p = p_0) = \sum_{k=c}^n \binom{n}{k} p_0^k (1 - p_0)^{n-k} \leq \alpha$$

Rejection Regions for the Exact Binomial Test

For the **upper-tailed** test $H_a : p > p_0$, we reject for large X :

The rejection region is $\{c, c + 1, \dots, n\}$ where c is chosen so that:

$$P(X \geq c \mid p = p_0) = \sum_{k=c}^n \binom{n}{k} p_0^k (1 - p_0)^{n-k} \leq \alpha$$

Complication with discrete distributions: Because X is discrete, we generally **cannot** find c such that $P(X \geq c) = \alpha$ exactly.

Rejection Regions for the Exact Binomial Test

For the **upper-tailed** test $H_a : p > p_0$, we reject for large X :

The rejection region is $\{c, c + 1, \dots, n\}$ where c is chosen so that:

$$P(X \geq c \mid p = p_0) = \sum_{k=c}^n \binom{n}{k} p_0^k (1 - p_0)^{n-k} \leq \alpha$$

Complication with discrete distributions: Because X is discrete, we generally **cannot** find c such that $P(X \geq c) = \alpha$ exactly.

Instead, we choose c as the **smallest** value such that $P(X \geq c \mid p_0) \leq \alpha$. This means the actual significance level is often *less than* α – the test is **conservative** (Devore 9.3).

Computing the Exact P-Value

The p-value for the exact binomial test is the probability, under H_0 , of observing a result as extreme or more extreme than the observed data:

Alternative	Exact P-value
$H_a : p > p_0$	$P(X \geq x_{obs} \mid p_0) = \sum_{k=x_{obs}}^n \binom{n}{k} p_0^k (1 - p_0)^{n-k}$
$H_a : p < p_0$	$P(X \leq x_{obs} \mid p_0) = \sum_{k=0}^{x_{obs}} \binom{n}{k} p_0^k (1 - p_0)^{n-k}$
$H_a : p \neq p_0$	$2 \cdot \min \{P(X \leq x_{obs}), P(X \geq x_{obs})\}$

Computing the Exact P-Value

The p-value for the exact binomial test is the probability, under H_0 , of observing a result as extreme or more extreme than the observed data:

Alternative	Exact P-value
$H_a : p > p_0$	$P(X \geq x_{obs} \mid p_0) = \sum_{k=x_{obs}}^n \binom{n}{k} p_0^k (1 - p_0)^{n-k}$
$H_a : p < p_0$	$P(X \leq x_{obs} \mid p_0) = \sum_{k=0}^{x_{obs}} \binom{n}{k} p_0^k (1 - p_0)^{n-k}$
$H_a : p \neq p_0$	$2 \cdot \min \{P(X \leq x_{obs}), P(X \geq x_{obs})\}$

In R, these binomial probabilities are straightforward to compute using `pbinom()`:

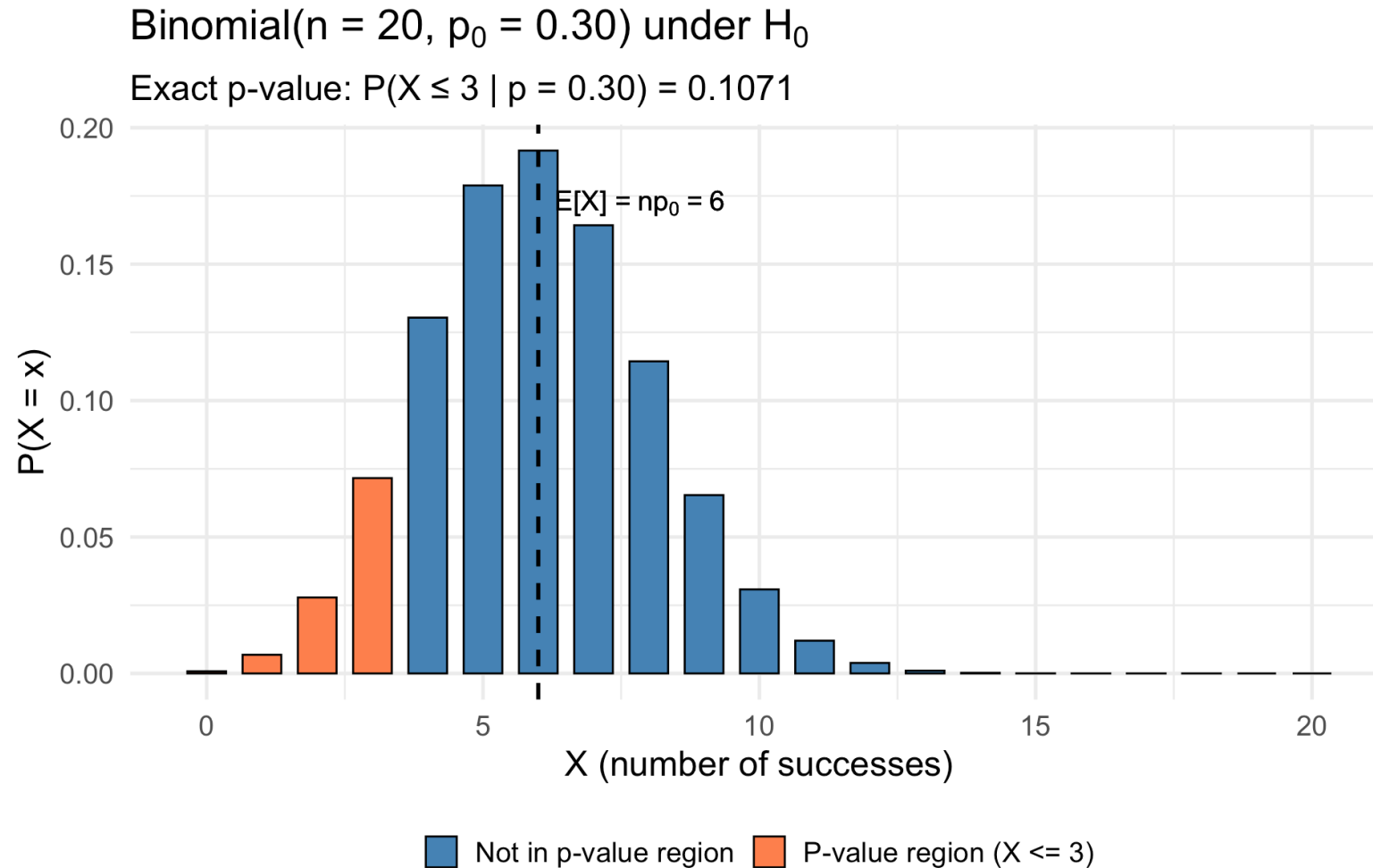
```
1 # Example: x = 3 successes out of n = 20, testing p0 = 0.30
2 x_ex <- 3; n_ex <- 20; p0_ex <- 0.30
3
4 # P(X <= 3 | p = 0.30) for lower-tailed test
5 p_lower <- pbinom(x_ex, size = n_ex, prob = p0_ex)
6
7 # P(X >= 3 | p = 0.30) for upper-tailed test
8 p_upper <- 1 - pbinom(x_ex - 1, size = n_ex, prob = p0_ex)
9
10 cat("P(X <= 3 | p=0.30) =", round(p_lower, 4), "\n")
```

P(X <= 3 | p=0.30) = 0.1071



Visualizing the Exact Binomial Distribution

► Code



Step-by-Step Exact Calculation

Let's compute $P(X \leq 3 \mid p_0 = 0.30, n = 20)$ term by term:

```
1 # Each individual probability
2 for (k in 0:x_ex) {
3   prob_k <- dbinom(k, size = n_ex, prob = p0_ex)
4   cat(sprintf("P(X = %d) = C(20,%d) × 0.30^%d × 0.70^%d = %.6f\n",
5               k, k, k, n_ex - k, prob_k))
6 }
```

$P(X = 0) = C(20,0) \times 0.30^0 \times 0.70^{20} = 0.000798$

$P(X = 1) = C(20,1) \times 0.30^1 \times 0.70^{19} = 0.006839$

$P(X = 2) = C(20,2) \times 0.30^2 \times 0.70^{18} = 0.027846$

$P(X = 3) = C(20,3) \times 0.30^3 \times 0.70^{17} = 0.071604$

```
1 cat("\n--- Sum ---\n")
```

--- Sum ---

```
1 cat("P(X <= 3) =", round(sum(dbinom(0:x_ex, n_ex, p0_ex)), 6), "\n\n")
```

$P(X \leq 3) = 0.107087$

```
1 cat("Since", round(p_lower, 4), "> 0.05, we fail to reject H0")
```

Since $0.1071 > 0.05$, we fail to reject H_0

Using `binom.test()` in R

```
1 # The easy way: binom.test() does all this for us
2 binom.test(x = 3, n = 20, p = 0.30, alternative = "less")
```

Exact binomial test

data: 3 and 20

number of successes = 3, number of trials = 20, p-value = 0.1071

alternative hypothesis: true probability of success is less than 0.3

95 percent confidence interval:

0.0000000 0.3436638

sample estimates:

probability of success
0.15

Understanding `binom.test()` Output

```
1 # Tidy output
2 binom_result <- binom.test(x = 3, n = 20, p = 0.30, alternative = "less")
3
4 cat("Test statistic (X):", binom_result$statistic, "\n")
```

Test statistic (X): 3

```
1 cat("Exact p-value:", round(binom_result$p.value, 4), "\n")
```

Exact p-value: 0.1071

```
1 cat("95% CI for p:", round(binom_result$conf.int[1], 4), ",",
2     round(binom_result$conf.int[2], 4), ")\n")
```

95% CI for p: (0 , 0.3437)

```
1 cat("Sample proportion (p-hat):", binom_result$estimate, "\n\n")
```

Sample proportion (p-hat): 0.15

```
1 # Verify the p-value matches our manual calculation
2 cat("Manual P(X <= 3 | p=0.30):", round(pbinom(3, 20, 0.30), 4))
```

Manual P(X <= 3 | p=0.30): 0.1071

Understanding `binom.test()` Output

```
1 # Tidy output
2 binom_result <- binom.test(x = 3, n = 20, p = 0.30, alternative = "less")
3
4 cat("Test statistic (X):", binom_result$statistic, "\n")
```

Test statistic (X): 3

```
1 cat("Exact p-value:", round(binom_result$p.value, 4), "\n")
```

Exact p-value: 0.1071

```
1 cat("95% CI for p:", round(binom_result$conf.int[1], 4), ",",
2     round(binom_result$conf.int[2], 4), ")\n")
```

95% CI for p: (0 , 0.3437)

```
1 cat("Sample proportion (p-hat):", binom_result$estimate, "\n\n")
```

Sample proportion (p-hat): 0.15

```
1 # Verify the p-value matches our manual calculation
2 cat("Manual P(X <= 3 | p=0.30):", round(pbinom(3, 20, 0.30), 4))
```

Manual P(X <= 3 | p=0.30): 0.1071

CI from `binom.test()`

The confidence interval from `binom.test()` is the **Clopper-Pearson** exact interval, which inverts the exact binomial test. It guarantees at least $100(1 - \alpha)\%$ coverage but can be conservative (wider than necessary). This is different from the Score (Wilson) interval returned by `prop.test()`.

Medical Example: Rare Adverse Drug Reaction

A pharmaceutical company is testing whether a new drug's rate of a serious adverse reaction exceeds the known background rate of 2%. In a Phase II trial, 4 out of 80 patients experience the reaction.

```
1 # Data
2 x_adr <- 4; n_adr <- 80; p0_adr <- 0.02
3
4 cat("p-hat =", x_adr / n_adr, "\n")
```

p-hat = 0.05

```
1 cat("Check np0 =", n_adr * p0_adr, "< 10: Normal approx NOT valid!\n\n")
```

Check np0 = 1.6 < 10: Normal approx NOT valid!

```
1 # Exact binomial test: H0: p = 0.02 vs Ha: p > 0.02
2 exact_result <- binom.test(x_adr, n_adr, p = p0_adr, alternative = "greater")
3 cat("Exact p-value:", round(exact_result$p.value, 4), "\n\n")
```

Exact p-value: 0.0769

```
1 # Compare with (invalid) z test for illustration
2 z_invalid <- (x_adr/n_adr - p0_adr) / sqrt(p0_adr * (1 - p0_adr) / n_adr)
3 p_z_invalid <- 1 - pnorm(z_invalid)
4 cat("Z test p-value (INVALID, for comparison):", round(p_z_invalid, 4), "\n")
```

Z test p-value (INVALID, for comparison): 0.0276

```
1 cat("Note the difference: exact and approximate p-values diverge\n")
```

Note the difference: exact and approximate p-values diverge

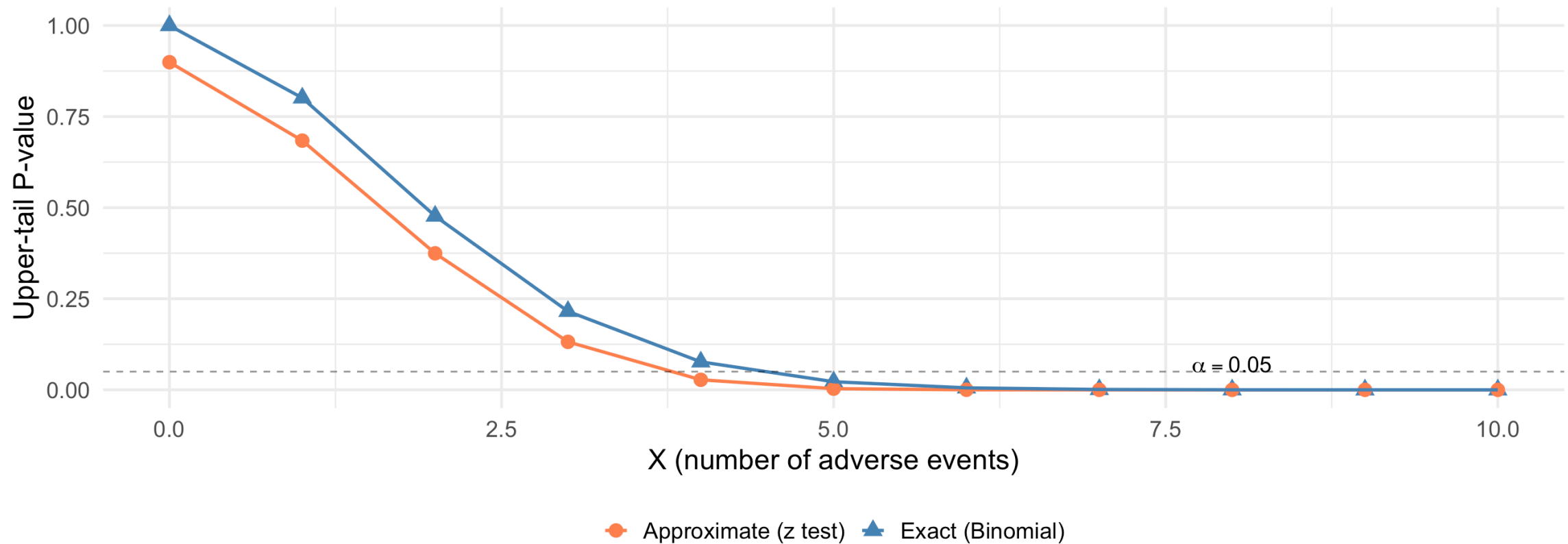


Exact vs Approximate: When Does It Matter?

► Code

Exact vs Approximate P-values: $n = 80$, $p_0 = 0.02$

Normal approximation unreliable when $np_0 = 1.6 < 10$



When to Use Each Test

Practical Guidelines for Proportion Tests

Use `prop.test()` (score/chi-squared test) when:

- $np_0 \geq 10$ and $n(1 - p_0) \geq 10$
- You want the Score (Wilson) confidence interval
- Consistency with Lesson 8's interval estimation framework

Use `binom.test()` (exact binomial) when:

- $np_0 < 10$ or $n(1 - p_0) < 10$ (small sample or rare event)
- You want guaranteed Type I error control
- The proportion is very close to 0 or 1
- You need the Clopper-Pearson exact CI

Devore Example 9.15: Obesity Prevalence

An earlier survey found 20% of Americans considered themselves obese. A recent study observed 686 obese individuals among 2014 adult men. Is the true obesity rate more than 1.5 times the self-assessment rate (i.e., more than 30%)?

```
1 # H0: p = 0.30 vs Ha: p > 0.30, alpha = 0.10
2 x_obs <- 686
3 n_obs <- 2014
4 p0_obs <- 0.30
5 alpha_obs <- 0.10
6
7 phat_obs <- x_obs / n_obs
8 se_obs <- sqrt(p0_obs * (1 - p0_obs) / n_obs)
9 z_obs <- (phat_obs - p0_obs) / se_obs
10
11 cat("p-hat =", round(phat_obs, 3), "\n")
```

p-hat = 0.341

```
1 cat("z =", round(z_obs, 2), "\n")
```

z = 3.98

```
1 cat("z_{0.10} =", round(qnorm(1 - alpha_obs), 3), "\n")
```

z_{0.10} = 1.282

```
1 cat("Decision:", ifelse(z_obs >= qnorm(1 - alpha_obs),
```

Example: Clinical Trial 30-Day Readmission (C&H 8.3 Style)

Setting: A hospital targets a 30-day readmission rate at or below 15%. Among 180 discharged patients, 35 were readmitted within 30 days.

```
1 # Data
2 x_readmit <- 35
3 n_readmit <- 180
4 p0_readmit <- 0.15
5 phat_readmit <- x_readmit / n_readmit
6
7 cat("p-hat =", round(phat_readmit, 4), "\n")
```

p-hat = 0.1944

```
1 cat("Check: np0 =", n_readmit * p0_readmit,
2     ", n(1-p0) =", n_readmit * (1 - p0_readmit), "\n\n")
```

Check: np0 = 27 , n(1-p0) = 153

```
1 # H0: p = 0.15 vs Ha: p > 0.15
2 # Manual z test (score test)
3 se_readmit <- sqrt(p0_readmit * (1 - p0_readmit) / n_readmit)
4 z_readmit <- (phat_readmit - p0_readmit) / se_readmit
5 cat("Manual z =", round(z_readmit, 3), "\n")
```

Manual z = 1.67

```
1 cat("Manual p-value =", round(1 - pnorm(z_readmit), 4), "\n")
```


Readmission: Comparing All Three Approaches

```
1 # Score test via prop.test()
2 score_result <- prop.test(x_readmit, n_readmit, p = p0_readmit,
3                           alternative = "greater", correct = FALSE)
4
5 # Exact binomial test
6 exact_result <- binom.test(x_readmit, n_readmit, p = p0_readmit,
7                             alternative = "greater")
8
9 cat("=== Comparison of Methods ===\n\n")
```

=== Comparison of Methods ===

```
1 cat("Manual z test:          p-value =", round(1 - pnorm(z_readmit), 4), "\n")
```

Manual z test: p-value = 0.0475

```
1 cat("prop.test (score):      p-value =", round(score_result$p.value, 4), "\n")
```

prop.test (score): p-value = 0.0475

```
1 cat("binom.test (exact):     p-value =", round(exact_result$p.value, 4), "\n\n")
```

binom.test (exact): p-value = 0.0624

```
1 cat("Score (Wilson) CI:      (", round(score_result$conf.int[1], 4), ",",
2     round(score_result$conf.int[2], 4), ")\n")
```

Score (Wilson) CI: (0.1506 , 1)

```
1 cat("Clopper-Pearson CI:      (", round(exact_result$conf.int[1], 4), ",",
```

Lesson 13 - Tests About a Mean and Proportion



Power and Sample Size for Proportion Tests

The formulas for $\beta(p')$ and required n parallel the z test for means:

$$n = \begin{cases} \left[\frac{z_{\alpha} \sqrt{p_0 q_0} + z_{\beta} \sqrt{p' q'}}{p' - p_0} \right]^2 & \text{one-tailed} \\ \left[\frac{z_{\alpha/2} \sqrt{p_0 q_0} + z_{\beta} \sqrt{p' q'}}{p' - p_0} \right]^2 & \text{two-tailed (approx)} \end{cases}$$

where $q_0 = 1 - p_0$ and $q' = 1 - p'$.

```
1 # How many patients to detect a readmission rate of 0.20 with 80% power?
2 p0 <- 0.15; p_alt <- 0.20; alpha <- 0.05; beta_target <- 0.20
3
4 n_prop <- ((qnorm(1 - alpha) * sqrt(p0 * (1 - p0)) +
5             qnorm(1 - beta_target) * sqrt(p_alt * (1 - p_alt))) /
6             (p_alt - p0))^2
7
8 cat("Required sample size:", ceiling(n_prop), "patients")
```

Required sample size: 342 patients

power.prop.test() in R

```
1 # R has a built-in function for proportion test sample size
2 power.prop.test(p1 = 0.15, p2 = 0.20, sig.level = 0.05,
3                 power = 0.80, alternative = "one.sided")
```

Two-sample comparison of proportions power calculation

```
      n = 713.0383
     p1 = 0.15
     p2 = 0.2
sig.level = 0.05
  power = 0.8
alternative = one.sided
```

NOTE: n is number in *each* group

power.prop.test() in R

```
1 # R has a built-in function for proportion test sample size
2 power.prop.test(p1 = 0.15, p2 = 0.20, sig.level = 0.05,
3                 power = 0.80, alternative = "one.sided")
```

Two-sample comparison of proportions power calculation

```
      n = 713.0383
     p1 = 0.15
     p2 = 0.2
sig.level = 0.05
  power = 0.8
alternative = one.sided
```

NOTE: n is number in *each* group

Note

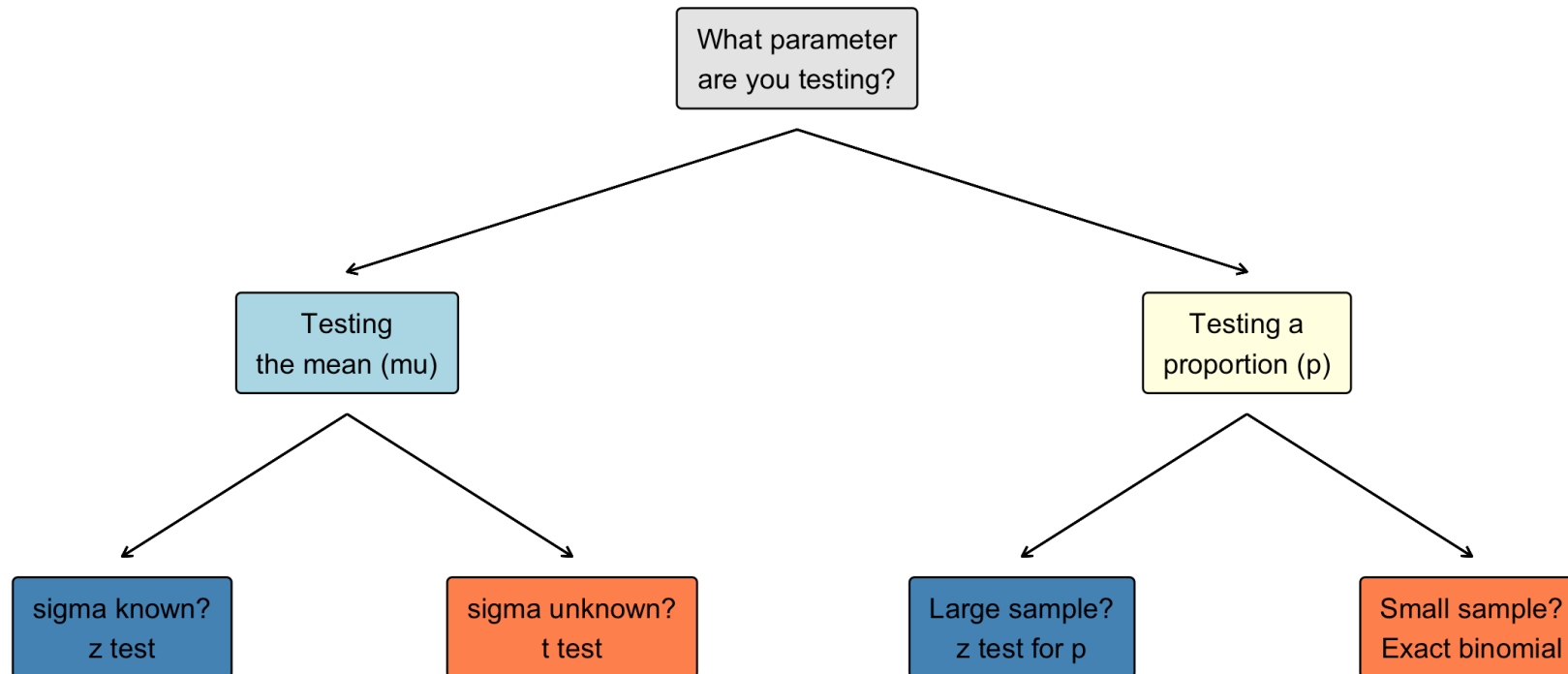
`power.prop.test()` is designed for **two-sample** tests by default. For one-sample tests, the sample size from our formula is a good approximation. For exact calculations, use the [pwr](#) package.

Putting It All Together

Decision Tree: Which Test to Use?

► Code

Decision Tree: Choosing the Right Test



Summary of Test Procedures

Test	Statistic	Null Distribution	R Function	Conditions
z test for μ	$\frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$	$N(0, 1)$	(manual)	σ known; normal pop. or large n
t test for μ	$\frac{\bar{x} - \mu_0}{s / \sqrt{n}}$	t_{n-1}	<code>t.test()</code>	σ unknown; normal pop. or large n
Score test for p	$\frac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$	$N(0, 1)$ or χ_1^2	<code>prop.test()</code>	$np_0 \geq 10$, $nq_0 \geq 10$
Exact binomial	X (count)	$\text{Binom}(n, p_0)$	<code>binom.test()</code>	Always valid

Summary of Test Procedures

Test	Statistic	Null Distribution	R Function	Conditions
z test for μ	$\frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$	$N(0, 1)$	(manual)	σ known; normal pop. or large n
t test for μ	$\frac{\bar{x} - \mu_0}{s / \sqrt{n}}$	t_{n-1}	<code>t.test()</code>	σ unknown; normal pop. or large n
Score test for p	$\frac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$	$N(0, 1)$ or χ_1^2	<code>prop.test()</code>	$np_0 \geq 10$, $nq_0 \geq 10$
Exact binomial	X (count)	$\text{Binom}(n, p_0)$	<code>binom.test()</code>	Always valid

Practical Advice

- In real medical research, σ is almost **never** known — use the **t test**
- The z test for means is primarily pedagogical (understanding the framework)
- For proportions, verify the validity condition before using the z test
- Always check assumptions (normality for small-sample t tests)

Complete Medical Example: Serum Creatinine Levels

A nephrology study examines whether a new drug changes serum creatinine levels. The normal reference is 1.0 mg/dL.

```
1 # Patient data
2 creatinine <- c(0.92, 1.15, 1.03, 0.88, 1.21, 0.97, 1.08, 1.14,
3               0.95, 1.06, 1.18, 0.99, 1.12, 0.91, 1.07, 1.01,
4               1.10, 0.94, 1.16, 1.03)
5
6 # Comprehensive analysis
7 cat("--- Summary Statistics ---\n")
```

--- Summary Statistics ---

```
1 cat("n =", length(creatinine), "\n")
```

n = 20

```
1 cat("x-bar =", round(mean(creatinine), 4), "\n")
```

x-bar = 1.045

```
1 cat("s =", round(sd(creatinine), 4), "\n")
```

s = 0.0976

```
1 cat("se =", round(sd(creatinine)/sqrt(length(creatinine)), 4), "\n")
```

se = 0.0218

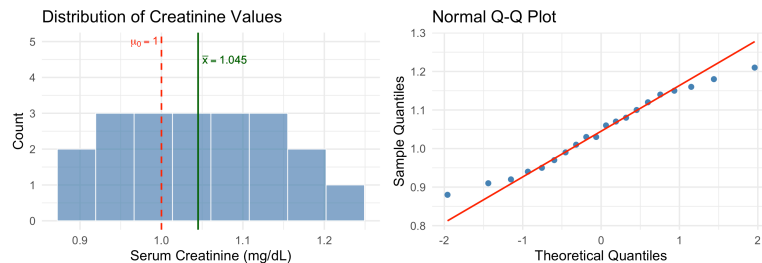
Serum Creatinine: Full Analysis

```
1 # Two-tailed t test:  $H_0: \mu = 1.0$  vs  $H_a: \mu \neq 1.0$ 
2 creat_test <- t.test(creatinine, mu = 1.0, alternative = "two.sided")
3 creat_test
```

One Sample t-test

```
data: creatinine
t = 2.0625, df = 19, p-value = 0.05311
alternative hypothesis: true mean is not equal to 1
95 percent confidence interval:
 0.9993331 1.0906669
sample estimates:
mean of x
 1.045
```

► Code



Serum Creatinine: Interpreting Results

```
1 # Summary of results
2 cat("=== Test Results ===\n")
```

=== Test Results ===

```
1 cat("t =", round(creat_test$statistic, 3), "\n")
```

t = 2.062

```
1 cat("df =", creat_test$parameter, "\n")
```

df = 19

```
1 cat("p-value =", round(creat_test$p.value, 4), "\n")
```

p-value = 0.0531

```
1 cat("95% CI: (", round(creat_test$conf.int[1], 4), ",",
2     round(creat_test$conf.int[2], 4), ")\n\n")
```

95% CI: (0.9993 , 1.0907)

```
1 if (creat_test$p.value < 0.05) {
2   cat("Interpretation: At alpha = 0.05, there IS statistically\n")
3   cat("significant evidence that the mean serum creatinine level\n")
4   cat("differs from the reference value of 1.0 mg/dL.\n\n")
5   cat("The 95% CI does NOT contain 1.0 -- consistent with rejection.")
6 } else {
7   cat("Interpretation: At alpha = 0.05, there is INSUFFICIENT\n")
8   cat("evidence that the mean serum creatinine level differs\n")
```



Simulation: Comparing z and t Tests

► Code

Simulated Type I error rates ($\alpha = 0.05$):

► Code

z test (sigma known): 0.0496

► Code

t test (sigma estimated): 0.051

► Code

Both correctly control Type I error at approximately 0.05

Key Takeaways and Looking Ahead

Key Takeaways

! Today's Main Points

1. **z test for μ :** Used when σ is known (rare in practice); test statistic $Z \sim N(0, 1)$ under H_0
2. **t test for μ :** The workhorse test when σ is unknown; test statistic $T \sim t_{n-1}$ under H_0
3. **Score test for p :** The z test for proportions is the score test; `prop.test()` computes $Z^2 \sim \chi_1^2$ and returns the Wilson/Score CI from Lesson 8
4. **Exact binomial:** Uses $P(X \leq x \mid p_0)$ directly from $\text{Binom}(n, p_0)$; always valid, especially when $np_0 < 10$
5. **CI-test duality:** A $100(1 - \alpha)\%$ CI contains all parameter values not rejected at level α

Key Takeaways

! Today's Main Points

1. **z test for μ :** Used when σ is known (rare in practice); test statistic $Z \sim N(0, 1)$ under H_0
2. **t test for μ :** The workhorse test when σ is unknown; test statistic $T \sim t_{n-1}$ under H_0
3. **Score test for p :** The z test for proportions is the score test; `prop.test()` computes $Z^2 \sim \chi_1^2$ and returns the Wilson/Score CI from Lesson 8
4. **Exact binomial:** Uses $P(X \leq x \mid p_0)$ directly from $\text{Binom}(n, p_0)$; always valid, especially when $np_0 < 10$
5. **CI-test duality:** A $100(1 - \alpha)\%$ CI contains all parameter values not rejected at level α

Common mistakes to avoid:

- Using $\hat{p}$ instead of p_0 in the SE for the proportion test (this would be Wald, not score!)
- Using `prop.test()` when $np_0 < 10$ — use `binom.test()` instead
- Applying the t test to very small samples from non-normal populations
- Saying “accept H_0 ” instead of “fail to reject H_0 ”

Looking Ahead

Next class (Lesson 14): P-Values; Power and Sample Size Determination (Devore 9.2, 9.24)

- P-values as a continuous measure of evidence
- Power curves for z and t tests
- Sample size determination for planning clinical trials
- The noncentral t distribution

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Sections 9.2–9.3
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Sections 8.1–8.3
- Mackowiak, P. A., Wasserman, S. S., & Levine, M. M. (1992). A critical appraisal of 98.6°F, the upper limit of the normal body temperature, and other legacies of Carl Reinhold August Wunderlich. *JAMA*, 268(12), 1578-1580.

Questions?

Thank you!